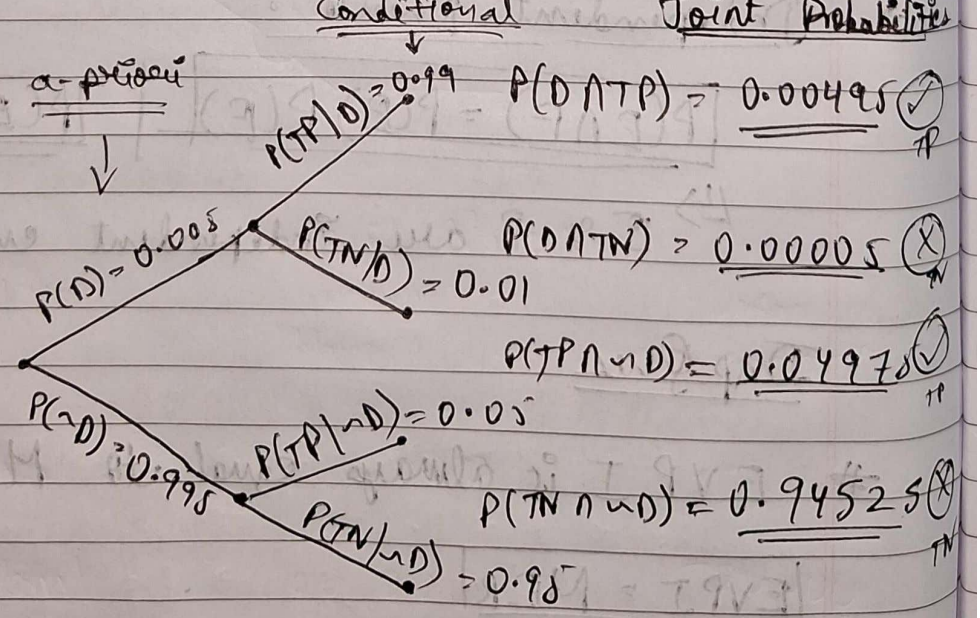
Decision Modeling
Week - 2

Bayes' Theorem :-

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

	If the disease is present	If the disease is not present
Probability of +ve result	0.99	0.05
Probability of -ve result	0.01	0.95

Probability of disease = 0.005



a-priori probabilities :- $P(D) = 0.005$

$P(\sim D) = 0.995$

Joint probabilities $P(D \cap TP) = P(TP|D)P(D)$

Total probability :-

$$P(TP) = P(D \cap TP) + P(\neg D \cap TP) = \underline{0.0547}$$

$$P(TN) = P(D \cap TN) + P(\neg D \cap TN) = \underline{0.9453}$$

Posterior probability :-

$$P(D|TP) = \frac{P(TP|D)\,P(D)}{P(TP)} = \frac{P(D \cap TP)}{P(TP)} = \underline{0.9095}$$

Probabilities :-

① Prior Probability :-
before evidence

F :- $\boxed{P(A)}$

② Joint Probability :-
both (or all) things happening

F :- $\boxed{P(A \cap B)}$

(Bayes)

③ Posterior Probability :-
after evidence

F :- $P(A|B) = \dfrac{P(B|A)\,P(A)}{P(B)}$

④ Total probability :-
all possible scenarios

F :- $P(A) = \sum_i P(A|B_i)P(B_i)$

(Normal)

⑤ Conditional Probability :-
"given" something has happened

F :- $P(A|B) = \dfrac{P(A \cap B)}{P(B)}$

→ Posterior probability is a special case of Conditional probability

Decision Trees :-

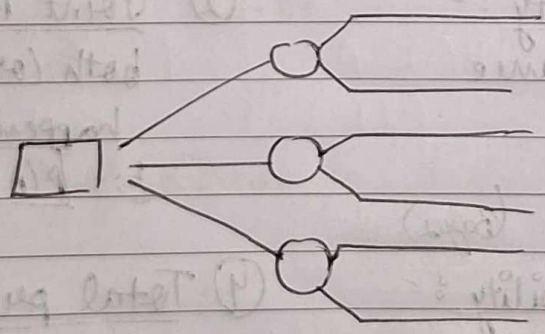
They represent decisions and outcomes in a sequential (chronological) order.

Symbols

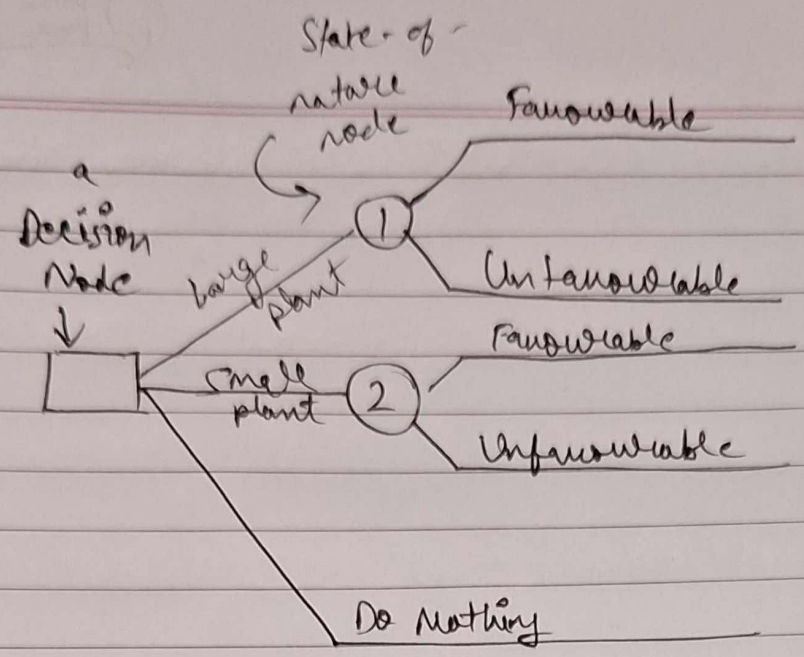
□ → a decision node from which one of several alternatives may be selected.

O → a state of nature node (chance node) out of which one state of nature will occur.

— → an arc (branch) represents possible decisions or states of nature.



Alternative	State of Nature: Favourable Market (0.5)	Unfavourable Market (0.5)
Large Plant	200,000	−180,000
Small Plant	100,000	−20,000
Do nothing	0	0



⊛ Expected Value of Sample Information (EVSI):-

→ Company wants to know the actual value of doing the survey.

→ Defined as the increase in expected value from the sample information resulting

$$EVSI = \left[\begin{array}{c}\text{Expected} \\ \text{Value of with} \\ \text{Sample info,} \\ \text{along with cost} \\ \text{to gather it}\end{array}\right] - \left[\begin{array}{c}\text{Expected value} \\ \text{of best decision} \\ \text{without the} \\ \text{sample info}\end{array}\right]$$

$$EVSI = \left(\begin{array}{c}\text{EV with} \\ \text{Sample info}\end{array} + \text{Cost}\right) - \left(\begin{array}{c}\text{EV without} \\ \text{Sample info}\end{array}\right)$$

⊛ Decision Making with Experimentation:-

⇒ Decision tree problem on plant construction with market survey.

⇓

experimentation.

(Look at this part from PPT)